IBP for Bubble

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Abstract

In this note we construct IBP relations for bubble-like cosmological correlators.

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Notations The metric reads: $\mathrm{d}s^2 = a^2(-\mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2)$, where the scale factor is $a(\tau) = -1/(H\tau)$. We set $H \equiv 1$ throughout this note.

1 Integrand

We define:

$$h_\nu^{(1)}(x) \equiv x^{-\nu}\mathrm{H}_\nu^{(1)}(x), \qquad h_\nu^{(2)}(x) \equiv x^{-\nu}\mathrm{H}_{\nu^*}^{(2)}(x), \qquad \nu \in \mathbb{C}, \quad x > 0. \tag{1}$$

It is easy to check that $h_\nu^{(1,2)}(x)$ are the two (linearly independent) solutions to:

$$h''(x) + \frac{2\nu+1}{x}h'(x) + h(x) = 0. \tag{2}$$

We can use $h_\nu^{(1,2)}(-k\tau)$ and power function of k and $-\tau$ to express the mode function and its conjugate of a scalar field with mass $m^2 = 9/4 - \nu^2$. For convenience, let us further define:

$$h_{\nu,0}(x) \equiv h_\nu(x), \qquad h_{\nu,n}(x) \equiv \frac{\mathrm{d}^n}{\mathrm{d}x^n}h(x), \tag{3}$$

It is straightforward that in general, the n-point correlator is the sum of several integrals as the following:

$$\int_{-\infty}^0 \mathrm{d}\tau_i \int \frac{\mathrm{d}^d\mathbf{k}_i}{(2\pi)^d}\frac{\mathcal{P}(\tau_i, p_i)}{\prod p_i^{n_i}} \times \prod e^{\pm \mathrm{i}P_i\tau_i} \times \prod h_n^{(1,2)}(-p_i\tau_i) \times \prod \theta(\tau_i - \tau_j). \tag{4}$$

Here $\boldsymbol{k}_i$ denotes the loop momenta, and $\boldsymbol{p}_i$ could be either loop momenta or external momenta, $p_i \equiv |\mathbf{p}_i|$, and P_i denotes the external energies.

2 Tadpole

Let us consider:

$$G[\{n_1, n_2\}, \{a\}, \{b\}] \equiv \int_{-\infty}^0 \mathrm{d}\tau \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3} e^{\mathrm{i}P\tau}\frac{(-\tau)^a}{|\boldsymbol{q}|^b} \times h_{n_1}(-q\tau)h_{n_2}(-q\tau). \tag{5}$$

2.1 Time IBP

$\int \partial_\tau[\cdots] = 0$ implies:

$$\begin{aligned} 0 =& \mathrm{i}PG[\{n_1, n_2\}, \{a\}, \{b\}] - aG[\{n_1, n_2\}, \{a-1\}, \{b\}] \\ &- G[\{n_1+1, n_2\}, \{a\}, \{b-1\}] - G[\{n_1, n_2+1\}, \{a\}, \{b-1\}]. \end{aligned} \tag{6}$$

or

$$\begin{aligned} 0 =& \mathrm{i}PG[\{n_1, n_2\}, \{a\}, \{b+1\}] - aG[\{n_1, n_2\}, \{a-1\}, \{b+1\}] \\ &- G[\{n_1+1, n_2\}, \{a\}, \{b\}] - G[\{n_1, n_2+1\}, \{a\}, \{b\}]. \end{aligned} \tag{7}$$

2.2 Momentum IBP

Using that

$$\boldsymbol{q}^i \partial_{\boldsymbol{q}^i} |\boldsymbol{q}|^b = \boldsymbol{q}^i \times b|\boldsymbol{q}|^{b-1} \times \partial_{\boldsymbol{q}^i}|\boldsymbol{q}| = \boldsymbol{q}^i \times b|\boldsymbol{q}|^{b-1} \times \frac{\boldsymbol{q}^i}{|\boldsymbol{q}|} = b|\boldsymbol{q}|^b, \tag{8}$$

we find $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots]$ implies: [ZQ: in general 3 should be spatial dimension d]:

$$\begin{aligned} 0 &= DG[\{n_1, n_2\}, \{a\}, \{b\}] - bG[\{n_1, n_2\}, \{a\}, \{b\}] \\ &+ G[\{n_1+1, n_2\}, \{a+1\}, \{b-1\}] + G[\{n_1, n_2+1\}, \{a+1\}, \{b-1\}]. \end{aligned} \tag{9}$$

or

$$\begin{aligned} 0 &= DG[\{n_1, n_2\}, \{a-1\}, \{b+1\}] - (b+1)G[\{n_1, n_2\}, \{a-1\}, \{b+1\}] \\ &+ G[\{n_1+1, n_2\}, \{a\}, \{b\}] + G[\{n_1, n_2+1\}, \{a\}, \{b\}]. \end{aligned} \tag{10}$$

Together with identity (**??**):

$$G[\{2, n_2\}, \{a\}, \{b\}] = -G[\{0, n_2\}, \{a\}, \{b\}] - (2\nu+1)G[\{1, n_2\}, \{a-1\}, \{b+1\}], \tag{11}$$

$$G[\{n_1, 2\}, \{a\}, \{b\}] = -G[\{n_1, 0\}, \{a\}, \{b\}] - (2\nu+1)G[\{n_1, 1\}, \{a-1\}, \{b+1\}], \tag{12}$$

we have completed the IBP relations for tadpoles.

Solving the ibp, we have

$$\mathrm{i}PG[\{n_1, n_2\}, \{a+1\}, \{b\}] + (2-b-a)G[\{n_1, n_2\}, \{a\}, \{b\}] = 0 \tag{13}$$

$$\partial_P G[\{n_1, n_2\}, \{a\}, \{b\}] = \frac{2-b-a}{P} G[\{n_1, n_2\}, \{a\}, \{b\}] \to \mathrm{d}G = (2-b-a)(\mathrm{d}\log P)G \tag{14}$$

3 Bubble

Let us consider:

$$\begin{aligned} G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] \equiv & \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3} e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s + \boldsymbol{q}|^{b_2}} \\ & \times h_{n_1}(-q\tau_1) h_{n_2}(-q\tau_2) h_{n_3}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau_1) h_{n_4}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau_2). \end{aligned} \tag{15}$$

From the Feynman rules, the natural $a_i + \nu_1 + \nu_2$ and $b_i + \nu_1 + \nu_2$ are half-integer or integer.

3.1 Time IBP

$\int \partial_{\tau_1}[\cdots] = 0$ implies:

$$\begin{aligned} 0 = & -\mathrm{i}P_1 G[\{n_1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1, b_2\}] - a_1 G[\{n_1, n_2, n_3, n_4\}, \{a_1-1, a_2\}, \{b_1, b_2\}] \\ & - G[\{n_1+1, n_2, n_3, n_4\}, \{a_1, a_2\}, \{b_1-1, b_2\}] - G[\{n_1, n_2, n_3+1, n_4\}, \{a_1, a_2\}, \{b_1, b_2-1\}]. \end{aligned} \tag{16}$$

Similarly, $\int \partial_{\tau_2}[\cdots] = 0$ implies:

$$
\begin{aligned}
0 = &- \mathrm{i}P_2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2-1\},\{b_1,b_2\}] \\
&- G[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2\},\{b_1,b_2-1\}].
\end{aligned} \tag{17}
$$

[CJQ: notice the property: if people use these 2 eqs for $b_i + 1 \to b_i$ (for k), we have $a_i \to a_i - 1$ (for τ). This is due to that k and τ is a pair of conjugate variables.

so we redefine b_i as $-b_i$ later, and I suggest to rewrite all b_i as $-b_i$ at the beginning.

also notice that after the redefinition, this 2 IBPs could be used as iterative reduction relation, for lowering the total power $a_1 + a_2 + b_1 + b_2$, and keep the total power and lowering arbitrary one of the indices (but raising others)]

3.2 Momentum IBP

Using that:

$$
\boldsymbol{q}^i \partial_{\boldsymbol{q}^i} |\boldsymbol{q}|^b = \boldsymbol{q}^i \times b|\boldsymbol{q}|^{b-1} \times \partial_{\boldsymbol{q}^i}|\boldsymbol{q}| = \boldsymbol{q}^i \times b|\boldsymbol{q}|^{b-1} \times \frac{\boldsymbol{q}^i}{|\boldsymbol{q}|} = b|\boldsymbol{q}|^b, \tag{18}
$$

$$
\boldsymbol{q}^i \partial_{\boldsymbol{q}^i} |\boldsymbol{q}+\boldsymbol{k}_s|^b = \boldsymbol{q}^i \times b|\boldsymbol{q}+\boldsymbol{k}_s|^{b-1} \times \frac{\boldsymbol{q}^i + \boldsymbol{k}_s^i}{|\boldsymbol{q}+\boldsymbol{k}_s|} = \frac{1}{2} b|\boldsymbol{q}+\boldsymbol{k}_s|^{b-2}(q^2 + |\boldsymbol{q}+\boldsymbol{k}_s|^2 - k_s^2), \tag{19}
$$

we find $\int \partial_{\boldsymbol{q}^i}[\boldsymbol{q}^i \cdots]$ implies: [ZQ: in general 3 should be spatial dimension d]

$$
\begin{aligned}
0 = &\, DG[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&- b_1 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&- \frac{1}{2} b_2 \Big(G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] + G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-2,b_2+2\}] \\
&- k_s^2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2+2\}] \Big) \\
&+ G[\{n_1+1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1-1,b_2\}] \\
&+ G[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2+1\},\{b_1-1,b_2\}] \\
&+ \frac{1}{2} \Big(G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2-1\}] + G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1+2,b_2+1\}] \\
&- k_s^2 G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2+1\}] \Big) \\
&+ \frac{1}{2} \Big(G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2-1\}] + G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1-2,b_2+1\}] \\
&- k_s^2 G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2+1\}] \Big).
\end{aligned} \tag{20}
$$

Similarly, $\int \partial_{\boldsymbol{q}^i}[(\boldsymbol{q}^i+\boldsymbol{k}_s^i)\cdots]$ implies $(\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\} \to \{n_3,n_4,n_1,n_2\},\{a_1,a_2\},\{b_2,b_1\})$:

$$
\begin{aligned}
&0 = DG[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&-\frac{1}{2}b_1\Big(G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] + G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1+2,b_2-2\}] \\
&-k_s^2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1+2,b_2\}]\Big) \\
&-b_2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&+\frac{1}{2}\Big(G[\{n_1+1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1-1,b_2\}] + G[\{n_1+1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1+1,b_2-2\}] \\
&-k_s^2 G[\{n_1+1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1+1,b_2\}]\Big) \\
&+\frac{1}{2}\Big(G[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2+1\},\{b_1-1,b_2\}] + G[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2+1\},\{b_1+1,b_2-2\}] \\
&-k_s^2 G[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2+1\},\{b_1+1,b_2\}]\Big) \\
&+G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2-1\}] \\
&+G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2-1\}].
\end{aligned} \tag{21}
$$

Together with identity (**??**):

$$
\begin{aligned}
G[\{2,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = &-G[\{0,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&-(2\nu+1)G[\{1,n_2,n_3,n_4\},\{a_1-1,a_2\},\{b_1+1,b_2\}], \quad (22)\\
G[\{n_1,2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = &-G[\{n_1,0,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&-(2\nu+1)G[\{n_1,1,n_3,n_4\},\{a_1,a_2-1\},\{b_1+1,b_2\}], \quad (23)\\
G[\{n_1,n_2,2,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = &-G[\{n_1,n_2,0,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \\
&-(2\nu+1)G[\{n_1,n_2,1,n_4\},\{a_1-1,a_2\},\{b_1,b_2+1\}], \quad (24)\\
G[\{n_1,n_2,n_3,2\},\{a_1,a_2\},\{b_1,b_2\}] = &-G[\{n_1,n_2,n_3,0\},\{a_1,a_2\},\{b_1,b_2\}] \\
&-(2\nu+1)G[\{n_1,n_2,n_3,1\},\{a_1,a_2-1\},\{b_1,b_2+1\}], \quad (25)
\end{aligned}
$$

we have completed the IBP relations for the (factorized) bubble.

3.3 symmetry of the IBPs

3.4 Indices of the IBPs

In the code, we redefine $G[\{\cdots\},\{a_1,a_2\},\{b_1,b_2\}]$, The new $G[\{\cdots\},\{a_1,a_2\},\{b_1,b_2\}]$ equals the original $G[\{\cdots\},a_0+a_1,a_0+a_2,-b_0-b_1,-b_0-b_2]$ Then, the a_1,a_2,b_1,b_2 appear in the 4 IBPs

$$
\begin{pmatrix}
\{-1,0\} & \{0\} & \{-1,0\} & \{-1,0\} \\
\{0\} & \{-1,0\} & \{-1,0\} & \{-1,0\} \\
\{0,1\} & \{0,1\} & \{-2\sim 0\} & \{-1\sim 2\} \\
\{0,1\} & \{0,1\} & \{-1\sim 2\} & \{-2\sim 0\}
\end{pmatrix} \tag{26}
$$

3.5 bubbles with G_{++}

Let us consider:

$$G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] \equiv \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, e^{-\mathrm{i}P_1\tau_1-\mathrm{i}P_2\tau_2} \frac{(-\tau_1)^{a_1}(-\tau_2)^{a_2}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s+\boldsymbol{q}|^{b_2}} \times h_{n_1}(-q\tau_1)\theta_{12}^{1,1} h_{n_2}(-q\tau_2) h_{n_3}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau_1)\theta_{12}^{2,2} h_{n_4}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau_2). \tag{27}$$

[CJQ: defination of the θ is needed] and define the remaining term:

$$R_1[\{n_3,n_4\},\{a\},\{b_1,b_2\}] \equiv \tag{28}$$

$$e^{\pi \mathrm{Im}\,[\nu_{12}]} \frac{4i}{\pi} \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, e^{-\mathrm{i}(P_1+P_2)\tau} \frac{(-\tau)^{a}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s+\boldsymbol{q}|^{b_2}} h_{n_3}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau)\theta_{12}^{1,1}(0) h_{n_4}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau) \tag{29}$$

$$R_2[\{n_1,n_2\},\{a\},\{b_1,b_2\}] \equiv \tag{30}$$

$$e^{\pi \mathrm{Im}\,[\nu_{34}]} \frac{4i}{\pi} \int_{-\infty}^{0} \mathrm{d}\tau \int \frac{\mathrm{d}^3\boldsymbol{q}}{(2\pi)^3}\, e^{-\mathrm{i}(P_1+P_2)\tau} \frac{(-\tau)^{a}}{|\boldsymbol{q}|^{b_1}|\boldsymbol{k}_s+\boldsymbol{q}|^{b_2}} h_{n_1}(-q\tau)\theta_{12}^{1,1}(0) h_{n_2}(-q\tau) \tag{31}$$

where ν_{12} labels the mass of the propagator labeled by n_1 and n_2. The k-IBP is the same as before. The τ -IBP becomes:

$$\begin{aligned} 0 = &-\mathrm{i}P_1 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_1 G[\{n_1,n_2,n_3,n_4\},\{a_1-1,a_2\},\{b_1,b_2\}] \\ &- G[\{n_1+1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - G[\{n_1,n_2,n_3+1,n_4\},\{a_1,a_2\},\{b_1,b_2-1\}] \end{aligned} \tag{32}$$

$$+\delta_{n_1+n_2,1}(-1)^{n_1+1} R_1[\{n_3,n_4\},\{a_1+a_2-2\nu_{12}-1\},\{b_1+2\nu_{12}+1,b_2\}] \tag{33}$$

$$+\delta_{n_3+n_4,1}(-1)^{n_3+1} R_2[\{n_1,n_2\},\{a_1+a_2-2\nu_{34}-1\},\{b_1,b_2+2\nu_{34}+1\}]. \tag{34}$$

and

$$\begin{aligned} 0 = &-\mathrm{i}P_2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - a_2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2-1\},\{b_1,b_2\}] \\ &- G[\{n_1,n_2+1,n_3,n_4\},\{a_1,a_2\},\{b_1-1,b_2\}] - G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2\},\{b_1,b_2-1\}] \end{aligned} \tag{35}$$

$$+\delta_{n_1+n_2,1}(-1)^{n_2+1} R_1[\{n_3,n_4\},\{a_1+a_2-2\nu_{12}-1\},\{b_1+2\nu_{12}+1,b_2\}] \tag{36}$$

$$+\delta_{n_3+n_4,1}(-1)^{n_4+1} R_2[\{n_3,n_4\},\{a_1+a_2-2\nu_{34}-1\},\{b_1,b_2+2\nu_{34}+1\}]. \tag{37}$$

. where we have use $\theta(0) = 1/2$

4 Differential equations for bubbles

Not hard to find

$$\begin{aligned} \partial_{P_1} G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] &= -iG[\{n_1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1,b_2\}] \\ \partial_{P_2} G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] &= -iG[\{n_1,n_2,n_3,n_4\},\{a_1,a_2+1\},\{b_1,b_2\}] \end{aligned} \tag{38}$$

If $P_1 = P_2 = P$, then

$$\begin{aligned}\partial_P G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = &- iG[\{n_1,n_2,n_3,n_4\},\{a_1+1,a_2\},\{b_1,b_2\}] \\ &- iG[\{n_1,n_2,n_3,n_4\},\{a_1,a_2+1\},\{b_1,b_2\}]\end{aligned} \tag{39}$$

For the K_s derivative, we need to consider

$$\partial_{k_s}|\boldsymbol{q}+\boldsymbol{k}_s| = \frac{k_s^2 + k_s\cdot q}{k_s|\boldsymbol{q}+\boldsymbol{k}_s|} = \frac{k_s^2 + |\boldsymbol{q}+\boldsymbol{k}_s|^2 - q^2}{2k_s|\boldsymbol{q}+\boldsymbol{k}_s|} \tag{40}$$

Then

$$\begin{aligned}&\partial_{k_s} G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] = -\frac{b_2}{2k_s}\Big(k_s^2 G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2+2\}] \\ &+ G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1,b_2\}] - G[\{n_1,n_2,n_3,n_4\},\{a_1,a_2\},\{b_1-2,b_2+2\}]\Big) \\ &+ \frac{1}{2k_s}\Big(k_s^2 G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2+1\}] \\ &+ G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1,b_2-1\}] - G[\{n_1,n_2,n_3+1,n_4\},\{a_1+1,a_2\},\{b_1-2,b_2+1\}]\Big) \\ &+ \frac{1}{2k_s}\Big(k_s^2 G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2+1\}] \\ &+ G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1,b_2-1\}] - G[\{n_1,n_2,n_3,n_4+1\},\{a_1,a_2+1\},\{b_1-2,b_2+1\}]\Big)\end{aligned} \tag{41}$$

for ∂_{k_s}, it acts on two parts: first, propagators $|q|$ and $|q+k_s|$; second, h functions. for the first part, even power term is the same as flat propagators, for example q^2. Thus, $\partial_{k_s^2}$ act on it is nearly the same as flat cases, which only yield even power terms. For h function,

$$\begin{aligned}\partial_{k_s} h(0;z) &\sim \tau h(1;z) \\ \partial_{k_s} h(1;z) &\sim \tau h(2;z)\end{aligned}$$

if we re-define $h = z^{1/2}H$ to satisfy

$$h''(z) + \left(1 + \frac{v^2 - 1/4}{z^2}\right) h = 0 \tag{42}$$

which do not involve odd power z terms. So, if we define the even or odd power in this system by the odevity of $a_1 + n_1 + n_2$, $a_2 + n_3 + n_4$, b_1 and b_2, then, l-IBP and ∂_{k_s} will not change the odevity of these four indices. On the contrary, in the original definition of h, one could also define the even-odd indices, but due to the h'/z term in the differential equation of h (typically, $z = |q|\tau$), the q power is changed by 1, thus the odevity of b_i is not invariant, only the odevity of $a_1 + a_2 + b_1 + b_2 + n_1 + n_2 + n_3 + n_4$ is kept. Thus, in the new definition, the odevity property is better.

5 Odevity

There are two ways to classify the master integrals:

1. For dq-IBP and dτ-IBP, $n_1+n_2+b_1$ odd (even) and $n_3+n_4+b_2$ odd (even), a_1 and a_2 arbitrary.

2. For dq-IBP, $n_1+n_3+a_1$ odd (even) and $n_2+n_4+a_2$ odd (even), b_1 and b_2 arbitrary.

The first classification is closed under the k_s differential equation, while the second classification is closed under the P_i differential equations. As P_i differential equations are simpler, we first consider the second one.

6 dim-shift relation

With generalized Sylvester's determinant identity, it is easy to prove

$$|\partial_{x_{ij}}|_{i,j\le n_1}|x_{ij}|^\gamma = |x_{ij}|_{i,j>n_1}\left(\prod_{i=0}^{n_1-1}\gamma+i\right)|x_{ij}|^{\gamma-1} \tag{43}$$

and for symmetric matrix $x_{ij}=x_{ji}$, it is easy to prove

$$\begin{aligned}
&|a_{ij}\partial_{x_{ij}}|_{i,j\le n_1}|x_{ij}|^\gamma = |x_{ij}|_{i,j>n_1}\left(\prod_{i=0}^{n_1-1}\gamma+\frac{i}{2}\right)|x_{ij}|^{\gamma-1}\\
&a_{ij}=\frac{1}{2}\ \text{ for }\ i\neq j,\quad a_{ij}=1\ \text{ for }\ i=j
\end{aligned} \tag{44}$$

Take

$$\begin{aligned}
&x_{ij}=q_i\cdot q_j,\quad n_1=L\\
&q_i=l_1,l_2,\cdots,l_L,p_1,p_2,\cdots,p_E.
\end{aligned} \tag{45}$$

(here the notation of x is not the same as sec2, be careful) we have $|x_{ij}|=\mathcal{G}$, $|x_{ij}|_{i,j>n_1}=\mathcal{K}$. So we have dim-raising relation:

$$\begin{aligned}
&C_{\gamma-1}\int Q\frac{\mathcal{G}^{\gamma-1}}{\mathcal{K}^{\gamma-1+L/2}}\prod dz_i\\
&=\frac{\left(\prod_{i=0}^{L-1}\gamma+\frac{i}{2}\right)C_\gamma}{C_{\gamma-1}}C_{\gamma-1}\int Q\frac{\mathcal{G}^{\gamma-1}}{\mathcal{K}^{\gamma-1+L/2}}\prod dz_i\\
&=C_\gamma\int Q|\partial_{x_{ij}}|_{i,j\le L}\frac{\mathcal{G}^\gamma}{\mathcal{K}^{\gamma+L/2}}\prod dz_i\\
&=C_\gamma\int(-1)^L Q|\overleftarrow{\partial}_{x_{ij}}|_{i,j\le L}\frac{\mathcal{G}^\gamma}{\mathcal{K}^{\gamma+L/2}}\prod dz_i\\
&Q=\frac{1}{\prod_i z_i^{a_i}},\quad \gamma=\frac{d-E-L-1}{2}
\end{aligned} \tag{46}$$

the left hand of this equation is d-2 dim integral, the right hand is the result after raising its dim.

7 canonical-like basis

For notational simplicity, in this section we shift the indices a and b, where a_i, b_i are integers.

$$G[\{\boldsymbol{n}\},\{a_0+a_1,a_0+a_2\},\{b_0+b_1,b_0+b_2\}] \to G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] \tag{47}$$

Define

$$z_1 = x_1^2 = q^2\,, \quad z_2 = x_2^2 = (q+k_s)^2\,. \tag{48}$$

The transformation for Baikov representation is

$$\begin{aligned}
&\mathrm{d}q^d \to C_d\, \mathcal{G}^{\frac{d-3}{2}} \mathcal{K}^{-\frac{d-2}{2}} \mathrm{d}z_1 \mathrm{d}z_2\,,\\
&\mathcal{G} = \frac{1}{4}\left(2z_1k_s^2 + 2z_2k_s^2 - k_s^4 - z_1^2 - z_2^2 + 2z_1z_2\right)\,, \quad \mathcal{K} = k_s^2\,, \quad C_d = \frac{\pi^{-1/2}}{\Gamma[(d-1)/2]}\,.
\end{aligned} \tag{49}$$

So, we have

$$G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1+1,b_2+1\}] \sim \cdots \times \frac{\mathrm{d}z_1}{x_1^{b_0+b_1+1}} \frac{\mathrm{d}z_2}{x_2^{b_0+b_2+1}} = \cdots \times 4 \frac{\mathrm{d}x_1}{x_1^{b_0+b_1}} \frac{\mathrm{d}x_2}{x_2^{b_0+b_2}} \tag{50}$$

By intersection theory, people have proved the differential equations for master integrals $\phi_i u$ is $\mathrm{d}\log$-form differential equations when ϕ is $\mathrm{d}\log$-form integrands and u satisfying $\mathrm{d}u = (\mathrm{d}\log u) * u$. Here we consider a generalization of that. In the framework of fibration intersection theory, we consider master integrals $\Phi_i.U$ with Φ_i and U are both vector. Choosing the U to satisfy $\mathrm{d}U = \mathrm{d}\Omega \cdot U$, and we construct the $\mathrm{d}\log$-form integrand Φ_i as master integral, we guess such selection of the master integral will satisfy $\mathrm{d}\log$-form differential equations.

Before define the U, let's notice that the DE of $\int \mathrm{d}\tau$ part, which is a 2-vertex tree-level-like integral family and has 2^4 master integrals and satisfy $\mathrm{d}\log$-form DE:

$$\begin{aligned}
I_{\boldsymbol{n}} \equiv & \int_{-\infty}^{0} \mathrm{d}\tau_1 \mathrm{d}\tau_2 e^{-\mathrm{i}P_1\tau_1 - \mathrm{i}P_2\tau_2} (-\tau_1)^{a_0} (-\tau_2)^{a_0}\\
& \times h_{n_1}(-q\tau_1) h_{n_2}(-q\tau_2) h_{n_3}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau_1) h_{n_4}(-|\boldsymbol{q}+\boldsymbol{k}_s|\tau_2).\\
\mathrm{d}I_{\boldsymbol{m}} =& (\mathrm{d}\Omega_{\boldsymbol{mn}}) I_{\boldsymbol{n}}\,, \qquad \boldsymbol{n} \equiv n_1, n_2, n_3, n_4\,,\\
\mathrm{d}\Omega =& \mathrm{d}\Omega_{x_1} + \mathrm{d}\Omega_{x_2} + \mathrm{d}\Omega_{ex}
\end{aligned} \tag{51}$$

where $\mathrm{d}\Omega_{x_i}$ is $\mathrm{d}\log$-form with only $\log x_i$, and Ω_{ex} is composed of $\mathrm{d}\log(P_i \pm x_1 \pm x_2)$ with constant coefficients, including ν and a_0.

Here we could select the U to be

$$\begin{aligned}
&U = x_1^{-b_0} x_2^{-b_0} \mathcal{G}^{-\epsilon} \mathcal{K}^{\epsilon} I_{\boldsymbol{n}}\,,\\
&\mathrm{d}U = -b_0(\mathrm{d}\log x_1 + \mathrm{d}\log x_2) - \epsilon(\mathrm{d}\log \mathcal{G} - \mathrm{d}\log \mathcal{K}) + \mathrm{d}\Omega
\end{aligned} \tag{52}$$

There are building blocks of $\mathrm{d}\log$-form integrand:

$$\begin{aligned}
&\mathrm{d}\log z_1 = 2\mathrm{d}\log x_1\,, \quad \mathrm{d}\log z_2 = 2\mathrm{d}\log x_2\,,\\
&\frac{\sqrt{z_2k_s^2}\mathrm{d}z_1}{\mathcal{G}} = -\mathrm{d}\log\left(\frac{2\sqrt{z_2k_s^2} + k_s^2 - z_1 + z_2}{-2\sqrt{z_2k_s^2} + k_s^2 - z_1 + z_2}\right)\,, \quad z_1 \leftrightarrow z_2\,.
\end{aligned} \tag{53}$$

Since $\mathrm{d}\int \mathrm{d}x_i/(P_i \pm x_1 \pm x_2)$ is a d log-form, we have another two building blocks of d log-form

$$\begin{aligned}
\partial_{P_j}\Omega_{ex}\mathrm{d}x_i &= \partial_{P_j}\Omega\mathrm{d}x_i \\
\partial_{x_j}\Omega_{ex}\mathrm{d}x_i &= (\partial_{x_j}\Omega - \partial_{x_j}\Omega_{x_j})\mathrm{d}x_i
\end{aligned} \tag{54}$$

Notice that

$$\begin{aligned}
G[\{\boldsymbol{n}\},\{a_1+1,a_2\},\{b_1,b_2\}] &= \partial_{P_1}G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] \\
&= (\partial_{P_1}\Omega_{\boldsymbol{nm}})\, G[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1,b_2\}]\,, \\
G[\{\boldsymbol{n}\},\{a_1,a_2\},\{b_1,b_2\}] &= \frac{1}{b_0+b_1-1}\,(\partial_{x_1}\Omega_{\boldsymbol{nm}})\, G[\{\boldsymbol{m}\},\{a_1,a_2\},\{b_1-1,b_2\}]\,.
\end{aligned} \tag{55}$$

we have

$$\begin{aligned}
\partial_{P_j}\Omega\mathrm{d}x_i &\sim \tau_j\mathrm{d}x_i\,, \\
\partial_{x_j}\Omega\mathrm{d}x_i &\sim \frac{b_0+b_1-1}{x_j}\mathrm{d}x_j\,.
\end{aligned} \tag{56}$$

could serves as a building block of d log-form integrand.

Construction of d log-form integrand :

$$k_s G[\{n_i\},\{0,0\},\{2,2\};d=3] \sim \frac{\mathrm{d}x_1\mathrm{d}x_2}{x_1x_2} \times x_1^{-b_0}x_2^{-b_0} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \tag{57}$$

$$k_s G[\{n_i\},\{1,0\},\{2,1\};d=3] \sim \frac{\mathrm{d}x_1}{x_1}\tau_1\mathrm{d}x_2 \times x_1^{-b_0}x_2^{-b_0} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \tag{58}$$

$$k_s G[\{n_i\},\{1,1\},\{1,1\};d=3] \sim \tau_1\mathrm{d}x_1\tau_2\mathrm{d}x_2 \times x_1^{-b_0}x_2^{-b_0} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}$$ [CJQ: wrong, double pole]

$$G[\{n_i\},\{0,0\},\{1,0\};d=1] \sim \frac{\mathrm{d}z_1}{z_1}\frac{k_s\sqrt{z_1}\mathrm{d}z_2}{\mathcal{G}} \times z_1^{-b_0/2}z_2^{-b_0/2} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \tag{59}$$

$$G[\{n_i\},\{1,0\},\{0,0\};d=1] \sim \tau_1\mathrm{d}x_1\frac{k_s\sqrt{z_1}\mathrm{d}z_2}{\mathcal{G}} \times z_1^{-b_0/2}z_2^{-b_0/2} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \tag{60}$$

$$(\partial_{x_2}\Omega_{ex}\mathrm{d}x_1)\frac{\mathrm{d}x_2}{x_2} \times x_1^{-b_0}x_2^{-b_0} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \tag{61}$$

$$(\partial_{x_2}\Omega_{ex}\mathrm{d}x_1)\frac{k_s\sqrt{z_1}\mathrm{d}z_2}{\mathcal{G}} \times z_1^{-b_0/2}z_2^{-b_0/2} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \tag{62}$$

To raise the dimension of $G[\{n_i\},\{a_1,a_2\},\{b_1,b_2\};d=1]$

$$\begin{aligned}
&G[\{n_i\},\{a_1,a_2\},\{b_1,b_2\};d=1] = \mathrm{d}z_1\mathrm{d}z_2 z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2} I_{\boldsymbol{n}}(-\tau_1)^{a_1}(-\tau_2)^{a_2}C_{1-2\epsilon}\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2} \\
&= \mathrm{d}z_1\mathrm{d}z_2 z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2} I_{\boldsymbol{n}}(-\tau_1)^{a_1}(-\tau_2)^{a_2}\partial_{q^2}\left[C_{3-2\epsilon}\mathcal{K}^{-\frac{1}{2}+\epsilon}\mathcal{G}^{-\epsilon}\right] \\
&= -\mathrm{d}z_1\mathrm{d}z_2 C_{3-2\epsilon}\mathcal{K}^{-\frac{1}{2}+\epsilon}\mathcal{G}^{-\epsilon}\partial_{q^2}\left[(-\tau_1)^{a_1}(-\tau_2)^{a_2}z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2} I_{\boldsymbol{n}}\right] \\
&= \mathrm{d}z_1\mathrm{d}z_2\left[\left(\frac{b_0+b_1}{2z_1}+\frac{b_0+b_2}{2z_2}\right)I_{\boldsymbol{n}} - (\partial_{z_1}+\partial_{z_2})\,I_{\boldsymbol{n}}\right] \\
&\quad\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}z_1^{-(b_0+b_1)/2}z_2^{-(b_0+b_2)/2}C_{3-2\epsilon}\mathcal{G}^{-\epsilon}\mathcal{K}^{-\frac{1}{2}+\epsilon} \\
&= 2\mathrm{d}x_1\mathrm{d}x_2\left[\left(\frac{b_0+b_1}{x_1^2}+\frac{b_0+b_2}{x_2^2}\right)I_{\boldsymbol{n}} - \left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right)I_{\boldsymbol{n}}\right] \\
&\quad\times(-\tau_1)^{a_1}(-\tau_2)^{a_2}x_1^{-(b_0+b_1-1)}x_2^{-(b_0+b_2-1)}C_{3-2\epsilon}\mathcal{G}^{-\epsilon}\mathcal{K}^{-\frac{1}{2}+\epsilon}
\end{aligned} \tag{63}$$

where we have used $\partial_{q^2} = \partial_{z_1} + \partial_{z_2} = \frac{1}{2x_1}\partial_{x_1} + \frac{1}{2x_2}\partial_{x_2}$. Then, we have

$$\begin{aligned}
&G[\{n_i\},\{0,0\},\{1,0\};d=1] \sim \mathrm{d}z_1\mathrm{d}z_2 z_1^{-b_0/2-1/2} z_2^{-b_0/2} I_{\boldsymbol{n}} \mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2} \\
&= 2\mathrm{d}x_1\mathrm{d}x_2 \left[\left(\frac{b_0+1}{x_1^2}+\frac{b_0}{x_2^2}\right) I_{\boldsymbol{n}} - \left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right) I_{\boldsymbol{n}}\right] x_1^{-b_0} x_2^{-b_0+1}\mathcal{G}^{-\epsilon}\mathcal{K}^{-\frac{1}{2}+\epsilon}
\end{aligned} \tag{64}$$

$$\begin{aligned}
&G[\{n_i\},\{1,0\},\{0,0\};d=1] \\
&= 2\mathrm{d}x_1\mathrm{d}x_2 \left[b_0\left(\frac{1}{x_1^2}+\frac{1}{x_2^2}\right) I_{\boldsymbol{n}} - \left(\frac{1}{x_1}\partial_{x_1}+\frac{1}{x_2}\partial_{x_2}\right) I_{\boldsymbol{n}}\right] (-\tau_1) x_1^{-b_0+1} x_2^{-b_0+1}\mathcal{G}^{-\epsilon}\mathcal{K}^{-\frac{1}{2}+\epsilon}
\end{aligned} \tag{65}$$

$$\begin{aligned}
&(\partial_{x_2}\Omega_{ex}\mathrm{d}x_1)\frac{\mathrm{d}x_2}{x_2} x_1^{-b_0} x_2^{-b_0} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\
=&(\partial_{x_2}\Omega - \partial_{x_2}\Omega_{x_2})\mathrm{d}x_1\frac{\mathrm{d}x_2}{x_2} x_1^{-b_0} x_2^{-b_0} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\
=&\mathrm{d}x_1\mathrm{d}x_2(\partial_{x_2} I_{\boldsymbol{n}}) x_1^{-b_0} x_2^{-b_0-1}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} - \mathrm{d}x_1\mathrm{d}x_2(\partial_{x_2}\Omega_{x_2}) x_1^{-b_0} x_2^{-b_0-1} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\
=&\mathrm{d}x_1(\mathrm{d}I_{\boldsymbol{n}}) x_1^{-b_0} x_2^{-b_0-1}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} - \mathrm{d}x_1\mathrm{d}x_2(\partial_{x_2}\Omega_{x_2}) x_1^{-b_0} x_2^{-b_0-1} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\
=&-\mathrm{d}x_1 I_{\boldsymbol{n}}\mathrm{d}\left(x_1^{-b_0} x_2^{-b_0-1}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon}\right) - \mathrm{d}x_1\mathrm{d}x_2(\partial_{x_2}\Omega_{x_2}) x_1^{-b_0} x_2^{-b_0-1} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\
=&\left(\frac{b_0+1}{x_2} - \partial_{x_2}\Omega_{x_2}\right) k_s x_1^{-b_0} x_2^{-b_0-1} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon-1/2}\mathrm{d}x_1\mathrm{d}x_2
\end{aligned} \tag{66}$$

$$\begin{aligned}
&(\partial_{x_2}\Omega_{ex}\mathrm{d}x_1)\frac{k_s\sqrt{z_1}\mathrm{d}z_2}{\mathcal{G}} \times z_1^{-b_0/2} z_2^{-b_0/2} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon}\mathcal{K}^{\epsilon} \\
=&\left(\frac{b_0-1}{x_2} - \partial_{x_2}\Omega_{x_2}\right) x_1^{-b_0+1} x_2^{-b_0+1} I_{\boldsymbol{n}}\mathcal{G}^{-\epsilon-1}\mathcal{K}^{\epsilon+1/2}\mathrm{d}x_1\mathrm{d}x_2
\end{aligned} \tag{67}$$